EXPERIMENT 17:

CODE:

```
% Sum of integers from 1 to N
sum(0, 0).

sum(N, S) :-
    N > 0,
    N1 is N - 1,
    sum(N1, S1),
    S is N + S1.
```

OUTPUT:

```
30 ?- consult(['c:/Users/Dhar/OneDrive/Desktop/Sum of integers from 1 to N.p1']).
true.

31 ?- sum(5, S).
S = 15
```

EXPERIMENT 18:

CODE:

File Edit View

```
% Database: Name and Date of Birth
person(dharshini, date(15, 8, 2004)).
person(priya, date(10, 3, 2005)).
person(rahul, date(20, 12, 2003)).
person(abi, date(5, 6, 2004)).
```

OUTPUT:

```
25 ?- consult(['c:/Users/Dhar/OneDrive/Desktop/% Database Name and Date of Birth.p1']).
true.

26 ?- student_teacher_subject(dharshini, Teacher, Subject).
ERROR: Unknown procedure: student_teacher_subject/3 (DWIM could not correct goal)
27 ?-  person(dharshini, DOB).
DOB = date(15, 8, 2004).
```

EXPERIMENT 19:

CODE:

File Edit View

```
% Student-Teacher-Subject Database
student_teacher_subject(dharshini, kumar, prolog).
student_teacher_subject(priya, ramesh, java).
student_teacher_subject(rahul, kumar, python).
student_teacher_subject(abi, priya, ai).
```

OUTPUT:

```
28 ?- consult(['c:/Users/Dhar/OneDrive/Desktop/% Student-Teacher-Subject Database.p1']).
true.

29 ?- student_teacher_subject(dharshini, Teacher, Subject).
Teacher = kumar,
Subject = prolog.
```

EXPERIMENT 20:

CODE:

File Edit View

```
% Planets Database
planet(mercury, 1).
planet(venus, 2).
planet(earth, 3).
planet(mars, 4).
planet(jupiter, 5).
planet(saturn, 6).
planet(uranus, 7).
planet(neptune, 8).
```

OUTPUT:

SWI-Prolog console

File Settings Tools Debug Help

```
Welcome to SWI-Prolog (threaded, 64 bits, version 10.0.2)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

21 ?- consult(['c:/Users/Dhar/OneDrive/Desktop/Planets Database.p1']).
true.

22 ?-  planet(Name, Number).
Name = mercury,
Number = 1 .

23 ?-  planet(earth, Position).
Position = 3.

24 ?-  planet(Name, 4).
Name = mars.

25 ?-
```